

Technical Specifications GPS ED167MP (OCXO-MQ)

RECEIVER:	6 channel C/A code receiver with external antenna/converter unit
ANTENNA:	Antenna/converter unit with remote power supply refer to chapter "Technical Specifications GPS167 Antenna"
ANTENNA INPUT:	antenna circuit dc-insulated; dielectric strength: 1000V= Length of cable: refer to chapter "Mounting the Antenna"
TIME TO SYNCHRO- NIZATION:	one minute with known receiver position and valid almanac 12 minutes if invalid battery buffered memory
PULSE OUTPUTS:	4 x PPS, change of second (P_SEC, 50 Ohm)
ACCURACY OF PULSES:	after synchronization and 20 minutes of operation: TCXO HQ/OCXO LQ : better than ± 250 nsec OCXO MQ/OCXO HQ : better than ± 100 nsec better than ± 2 μ sec during the first 20 minutes of operation
FREQUENCY OUTPUTS:	4 x 10 MHz, 50 Ohm
ACCURACY OF FREQUENCY:	$\pm 5 \times 10^{-12}$ /24h (see MQ-Oscillator specification)
FREQUENCY SYNTHESIZER:	10 MHz

SYNTHESIZER

OUTPUTS:	F_SYNTH:	TTL level	
	F_SYNTH_OD:	open drain	
		drain voltage:	< 100 V
		sink current to GND:	< 100 mA
		dissipation power at 25°C:	< 360 mW
	F_SYNTH_SIN	sine-wave	
		output voltage:	1.5 V eff.
		output impedance:	200 Ohm

TIME_SYN

OUTPUT: TTL HIGH level if synchronized

SERIAL PORTS 2 x asynchronous serial ports (RS-232)

Baud Rate: 300 up to 19200
Framing: 7N2, 7E1, 7E2, 8N1, 8N2, 8E1

default setting: COM0: 19200, 8N1
COM1: 9600, 8N1

Annotation: Even if one of the setup functions "INIT USER
PARMS" or "Resetting Factory Defaults" is
executed, the serial port parameters are
reset to default values only if invalid para
meters have been configured.

ETHERNET: 10/100 - TCP/IP

TIME CAPTURE

INPUTS: triggered on falling TTL slope
Interval of events: 1.5msec min.
Resolution: 100ns

POWER

REQUIREMENTS: 100-240 V.AC - 150 mA

PHYSICAL

DIMENSIONS: 19" module in a closed Aluminium case

FRONT PANEL: 3U / 21HP (128mm high x 107mm wide), Aluminium

REAR EDGE

CONNECTOR: according to DIN 41612, type C 64, rows a+c (male)

RF CONNECTOR: coaxial N and BNC (female)

AMBIENT

TEMPERATURE: 0 ... 50°C

HUMIDITY: 85% max.

Specifications of Internal Oscillator

	TCXO	OCXO LQ	OCXO MQ	
Kurzzeitstabilität $\tau = 1 \text{ sec}$	$2 * 10^{-9}$	$1 * 10^{-9}$	$2 * 10^{-10}$	
Freilaufgenauigkeit ein Tag	$\pm 1 * 10^{-7}$ ± 1 Hz (Note 1)	$\pm 2 * 10^{-8}$ ± 0.2 Hz (Note 1)	$\pm 1,5 * 10^{-9}$ ± 15 mHz (Note 1)	
Freilaufgenauigkeit ein Jahr	$\pm 1 * 10^{-6}$ ± 10 Hz (Note 1)	$\pm 4 * 10^{-7}$ ± 4 Hz (Note 1)	$\pm 1 * 10^{-7}$ ± 1 Hz (Note 1)	
Genauigkeit GPS-synchron 24 Std. gemittelt	$\pm 1 * 10^{-11}$	$\pm 1 * 10^{-11}$	$\pm 5 * 10^{-12}$	←
Phasenrauschen	1 Hz -60 dBc/Hz 10 Hz -90 dBc/Hz 100 Hz -120 dBc/Hz 1 kHz -130 dBc/Hz	1 Hz -60 dBc/Hz 10 Hz -90 dBc/Hz 100 Hz -120 dBc/Hz 1 kHz -130 dBc/Hz	1 Hz -75 dBc/Hz 10 Hz -110 dBc/Hz 100 Hz -130 dBc/Hz 1 kHz -140 dBc/Hz	
Stromversorgung bei 25°C Normalbetrieb Aufheizphase	+5V / 20 mA N/A	+5V / 160 mA +5V / 380 mA	+5V / 90 mA +5V / 330 mA	
Temperaturdrift im Freilauf	$\pm 1 * 10^{-6}$ (-20...70°C)	$\pm 2 * 10^{-7}$ (0...60°C)	$\pm 5 * 10^{-8}$ (-20...70°C)	

Note 1:

Die berechnete Genauigkeit in Hertz basiert auf der Normalfrequenz von 10 MHz. Zum Beispiel:

Genauigkeit des TCXO HQ (Freilauf ein Tag) ist $\pm 1 * 10^{-7} * 10 \text{ MHz} = \pm 1 \text{ Hz}$

Technical Specifications GPS167 Antenna

ANTENNA: dielectrical patch antenna, 25 x 25mm
receive frequency: 1575.42 MHz
bandwidth: 9 MHz

CONVERTER: local oscillator to converter frequency: 10 MHz
first IF frequency: 35.4 MHz

POWER

REQUIREMENTS: 12V ... 18V, @ 100mA (provided via antenna cable)

CONNECTOR: coax type N, female

AMBIENT

TEMPERATURE: -40 ... +65°C

HOUSING: ABS plastic case for outdoor installation (IP56)

PHYSICAL

DIMENSION:

